🔍 Advanced Computer Forensics Topics: Encoding, Hashing, and Hex Analysis

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🔐 1. Encoding and Encryption

These two terms are **often confused**, but they serve **different purposes**.

Aspect	Encoding	Encryption
Purpose	Convert data for compatibility	Hide data for confidentiality
Reversible?	Yes (with the correct scheme)	Yes (only with the key)
Examples	Base64, ASCII, Unicode	AES, RSA, DES
Use Case	Email attachments, URL transmission	Secure communications, data protection

- **Encoding**: Makes data readable by systems (not secure).

- **Encryption**: Secures data from unauthorized access.

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2. The Hex Editor

A **Hex Editor** allows forensic analysts to:

- View and edit **raw binary data** in hexadecimal format.
- Examine **file headers**, metadata, and anomalies.
- Recover hidden/deleted data, identify embedded files.

Example Tools:

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HxD

- WinHex

- Hex Fiend

- 010 Editor

Common Uses:

- File signature verification

- Disk analysis

- Manual data carving

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Malware examination

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3. Files (Forensic Relevance)

In forensics, files are:

- The **primary source** of evidence.
- Analyzed for **metadata**, content, and structure.
- Can contain **hidden payloads**, **steganography**, or **embedded malware**.

Analysis Techniques:

- **Header/Footer signature matching**

- **File carving**
- **Timestamp validation**
- **Comparison with known good copies**

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4. Hashing

Hashing is the process of generating a **fixed-length output (hash value)** from input data of any length.

Characteristics:

- One-way (cannot retrieve original data)
- Used for **integrity verification**

- Even a **1-bit change** causes completely different hash

Algorithm	Output Length
MD5	128-bit
SHA-1	160-bit
SHA-256	256-bit
SHA-3	224–512 bits

Uses:

- File integrity checking
- Password storage
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- Digital signatures
- Forensic evidence verification

🔗 5. Hashing DLs (Download Links)

Some websites provide **hashes** (MD5, SHA-256) for downloadable files.

Why?

- To verify **file integrity** after download
- Ensure the file has **not been tampered** with
- Prevent **man-in-the-middle attacks** or corruption

Forensic Use:

- Analysts verify hashes of downloaded evidence/toolkits
- Compare with **official hash** provided by trusted sources

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6. MD5 Hash Collisions

An **MD5 collision** occurs when **two different inputs** produce the **same hash value**.

Problems:

- Can be used to **forge digital certificates or signatures**
- **Reduces trust** in MD5 for cryptographic integrity
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Known exploits like **Flame malware** abused this

Forensics Response:

- **Avoid MD5 for authentication**
- Prefer **SHA-256 or higher** for integrity checks

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7. Hash Collisions

A **hash collision** is a situation where **two different inputs** result in the **same hash**.

Why It Matters:

- Undermines **trust in data integrity**
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Legal cases can be challenged if weak hashes are used

- Increases risk of **tampering**

Resistant Algorithms:

- SHA-256 (currently considered collision-resistant)
- SHA-3 (stronger and newer)

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8. Bit Rot (Data Degradation)

Bit rot, also called **data decay**, refers to the **gradual corruption of data stored on storage media** over time.

Causes:

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- Aging of storage (hard drives, SSDs, CDs, DVDs)
- Radiation, magnetism, manufacturing defects
- Inadequate **error correction codes (ECC)**

Impact in Forensics:

- May result in **file corruption**
- **Hash values mismatch**
- Recovery efforts may be needed (using tools like TestDisk, ddrescue)

Prevention:

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Regular backups

- ECC-enabled systems

- Data integrity verification using **checksums/hashes**